

AJMAN UNIVERSITY

College of Engineering and Information Technology

Electrical and Computer Engineering Department

COE431 – Project I

HydroGuard
(Smart Water Quality Bottle)

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PROJECT DECLARATIONS

1. It is declared that this project is original and is resulted as a group effort of the project members only, unless otherwise specified and in the document and references. Moreover, this work in its entirety is never submitted in any form to quality for another degree or diploma at any university/institution and I/We undertake not to submit the same in future for qualifying and obtaining any other degree from this or any other university/institution.

2. I/We affirm to have applied ethics to the design process and in the selection of the project design. I/We also affirm to have considered the safety of the public to be paramount and have addressed this in the design wherever applicable.

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LIST OF DESIGN CONSIDERATIONS

Please fill in the following table as per the design considerations of the project.

The impact of the project on:	Aspects	Applicability		Extent of Applicability				
Economy	The project covers feasibility of development, production and operational cost.	Yes	No	1	2	3	4	5
Environment	The project has an effect on environment (negative to positive)	Yes	No	1	2	3	4	5
Ethic	The project does not have any obvious potential conflicts with a professional Code of Ethics arising from the development or use of the product	Yes	No	1	2	3	4	5
Health and Safety	Safety and health of public is being considered while designing and developing the project and it does not violate any existing laws of health and safety	Yes	No	1	2	3	4	5
Society	The project is beneficial for the community and society as a whole. The impact on the society is considered	Yes	No	1	2	3	4	5

LIST OF STANDARDS FOLLOWED IN THE PROJECT

Please provide a list of standards followed in the design and implementation of the project

S. No.	Standard Name	Standard Body
1	**IEEE 802.11™**: Wireless LAN Medium Access Control (MAC) and Physical Layer (PHY) Specifications for Wi-Fi communication	IEEE
2	**IEEE 802.15.1™**: Bluetooth wireless communication standard (BR/EDR and BLE)	IEEE
3	**ISO 5667**: Water quality — Sampling methods	ISO
4	**ISO/IEC 25010**: Systems and Software Quality Requirements and Evaluation (SQuaRE)	ISO/IEC
5	**ISO/IEC 27001**: Information security management for cloud data handling	ISO/IEC
6	**WHO Guidelines for Drinking-water Quality**: Thresholds for contaminants and water safety classification	World Health Organization (WHO)
7	**I²C and UART Communication Protocol Standards**: For sensor–microcontroller interfacing	NXP / Industry Standard
8	**RoHS Directive (2011/65/EU)**: Restriction of Hazardous Substances in electrical and electronic equipment	European Union (EU)
9	**SI Units**: Standard units for pH, NTU, mg/L, mS/cm, °C in measurement reporting	Bureau International des Poids et Mesures (BIPM)

CERTIFICATE OF APPROVAL

This is to certify that the project titled "_____" accomplished by_____, Reg. ID. _____, under the supervision of Dr._____, Ajman University, UAE, is fully acceptable, in quality and scope, as ProjectX as required by the study plan of the degree of *B.Sc. in Computer Engineering*.

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ABSTRACT

Clean and safe drinking water is one of the basic human needs and millions of people across the world struggle every day to sustain its quality. The Smart Water Quality Testing Bottle is a portable, smart, and the AI-powered gadget that aims to deliver real-time water safety testing in urban as well as remote areas. This project combines a multi-sensor array (measuring PH, turbidity, total dissolved solids (TDS), electrical conductivity, temperature and dissolved oxygen) in a small form bottle-shaped form using an ESP32 microcontroller. On-device classification of safe or unsafe water based on the World Health Organization (WHO) guidelines is achieved using an embedded artificial intelligence system, which allows such system to continue operating successfully in offline mode not requiring cloud processing. The mobile application can be developed with the use of Blynk IoT system, enabling the real-time monitoring, visualization of historical data and push notifications in case of unsafe readings. Calibration standards give assurances of accuracy, whereas power-efficient hardware and rechargeable battery helps to operate longer in the field. Smart Water Quality Testing Bottle overcomes the drawbacks of the current products in the market as it provides multi-parameter sensing technology, artificial intelligence to interpret the sensor values, portability, and ease of usage, thus it can be used in households, outdoor activities, disaster relief, and environmental monitoring.

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CHAPTER 1

INTRODUCTION

Chapter 1: INTRODUCTION TO PROJECT

1.1 Overview

Clean drinking water is a basic human right, but millions of individuals are struggling to ensure that their water is safe daily in this world. It is under such urgency that this project proposes the Smart Water Quality Testing Bottle, which is innovative, compact, portable, and aims to test the safety/quality of drinking water in a real-time manner.

The system combines several water quality sensors along with a display on the board, AI algorithms and wireless communications into a slick bottle-type design. It is constructed to target people and communities that need a fast, precise, and convenient technique to determine water safety wherever they are. This report summarizes the development, application, and capabilities of this solution, which brings together embedded systems, artificial intelligence, and mobile communication to provide a potent and useful instrument of water sensing [19].

1.2 Statement of Problem

The problem of water pollution is still of concern all over the world and health problems are the most common conditions for people to get Ebola diarrhea, cholera etc. caused by water pollution [19]. Although laboratory checks are accurate, they are expensive, time consuming and not accessible to use on a daily or on-the-go basis. Moreover, most households, travelers and campers do not have equipment and knowledge to test water quality prior to their consumption.

Conventional handheld test kits consist of one to two parameters, can only be interpreted manually and do not provide digital reports. The issue is that there is no compact multi-sensor system available that would be able to:

- Multiparameter water analysis Test
- Show or broadcast the results on-demand
- Increase accuracy and interpretation with the help of AI
- Be cheap, portable and newbie-friendly

This project will be able to fill that gap by creating an intelligent water testing bottle that will be able to use several sensors and intelligent algorithms to recognize tainted water and alert the user in addition to capturing the data to be saved and retrieved either later or remotely.

1.3 Project Specifications

Smart Water Quality Testing Bottle grins with the following prime characteristics:

- 6-7 Sensors: such as pH, turbidity, TDS (Total Dissolved Solids), temperature, conductivity and maybe heavy metal sensing.
- Display Screen: Small internal monitor to indicate real-time readings and safety condition.
- Rechargeable Battery: field battery.
- Cup/Reservoir: It is used to sample water to be test upon.
- Artificial intelligence: Data about sensor patterns are used to train machines learning models to forecast water safety with higher accuracy.
- Connection: Bluetooth or Wi-Fi chip to sync a smartphone application or cloud dashboard.
- Mobile/Web App: An application that will display past records, send notifications and access water quality analytics in real time.
- Compact: All features embedded in a bottle-like case which is portable and easy-to-use.

1.4 Motivation/Justification

The primary drive in having such a project is to fill in the gap of the available water testing devices that are not readily open to the public and that they can utilize. Concerned, people require an easy and fast mechanism to know when the water is safe to ensure that diseases are not encountered in the countryside, on the road or even in cases of post-disaster assistance.

Existing water test machines are either:

- Laboratory and costly, or
- Manual and non-automatic Incomplete

This project democratizes water safety testing, by combination of cheap microcontrollers, smart sensors, and edge AI. It is especially helpful at:

- Inhabitants of areas that have unmet water supplies
- Red cross operations Here, the red cross is supposed to provide emergency relief teams to assist them.
- Out bounders, Hikers and Campers
- Environ mental education

1.5 Relevance to Sustainable Development Goals (SDGs)

This project directly supports the **United Nations Sustainable Development Goals (SDGs)**, particularly:

- **SDG 6: Clean Water and Sanitation**

  This project facilitates the vision of ensuring availability and sustainable management of water to all by creating a smart, portable, and cost-effective solution that monitors the quality of water. Real time capacity of the system to detect contaminants in the water supply like pH out of balance, high turbidity, unsafe TDS level, and unusual conductivity helps ensure drinking water quality. This can enhance good health, sanitation, and responsible water management among the developed and developing countries [19].

- **SDG 3: Good Health and Well-Being**

  Access to clean water is directly associated with elimination of water borne diseases like cholera, diarrhea and typhoid. The project guarantees reduced health risks by affording direct water safety classification and alert so that the world goal of healthy lives and promotion of well-being could be achieved globally [19].

- **SDG 9: Industry, Innovation, and Infrastructure**

  This system achieves innovation in environmental monitoring with merging of AI, IoT and multi-sensors in a small form factor. It promotes the implementation of intelligent technologies in infrastructure construction and sustainability of the environment [22].

By aligning with these goals, the Smart Water Quality Testing Bottle not only serves as a technological innovation but also as a tool for global sustainable development and public health improvement.

1.6 Project Applications

There are lots of applications for the Smart Water Quality Testing Bottle:

- Home Use: To determine the quality of tap or well water by the family.
- Outdoor: Hikers, campers or outdoors travelers to wild locations.
- Disaster Response: The areas that have been affected by floods or earthquake and water pollution are very prevalent.
- Undeveloped Regions: It can be utilized by the NGOs to confirm water sources in underserved or rural regions.
- Smart Cities: Incorporation into IoT systems of urban water quality testing.
- Research & Education: To study environment and create awareness through school/ college programs.

1.7 Discussion on Standards and Constraints

Engineering Standards Overview:

- **ISO 5667**: Standards for water quality sampling.
- **ISO/IEC 25010**: For software/system quality (reliability, usability, etc.)
- **IEEE 802.11:** Incorporated for enabling wireless communication [8].
- **ISO/IEC 27001**: If cloud storage is used, for securing user data.
- **I2C / UART Protocols**: For sensor communication.
- **WHO Guidelines for Drinking Water Quality**: Used as reference thresholds [20].

Design Constraints Overview:

It is through some practical and technical limitations that the design of the Smart Water Quality Testing Bottle has been determined, serving to strategically enforce its effectiveness, usability and cost. Given that the system is to be operated as portable apparatus using batteries, power conservation is of utmost importance so as to have long operations before admitting the need to recharge the batteries. Calibration of the sensors is also essential because the precision of the measurements should be observed

over a wide range of environmental settings and through long-term operation. Form factor and size also present a limitation, since all hardware must be packaged into a small, bottle-shaped package that does not reduce performance. It must also be affordable; the price of its production must be kept low so that the device can be affordable to the highest number of users, especially in the developing populations [21]. Also, it must be waterproof to avoid water splashes or other related damages to fragile electronics in the field under different conditions, and at the same time, to survive such accidents. Lastly, as the system uses AI-based analysis, the target level of accuracy can be only reached after training the model on a variety and sufficiently representative data set so that the safety classification can ensured in any situation in practice.

1.8 Organization of the Report

There is the first chapter that serves as a basis to the Smart Water Quality Testing Bottle by giving an overview of the project. This paper starts by giving the significance of the issue of water quality on human health and safety and then proceeds to the gist of the problem that this project will address. The chapter describes the specifications of the system, motivation of the project and identifies the applications of the system and beneficiaries in the real world. It also talks about the major engineering standards to be considered when designing the product, physical and technological constraints, that influenced work on the system.

The second chapter immerses into the literature review, with the emphasis on the state of the water quality monitoring technologies. It includes the following associated technologies: multi-parameter sensors, Internet of Things (IoT) based devices, AI in environmental sensing, and small embedded systems. Review of existing projects, commercial products and scholarly work is also given in the chapter and conclusions based on those were analyzed and research gaps that the current project attempts to bridge were established. This part forms the technical background and theoretical support to the design decisions that will be provided in the forthcoming sections of the report.

The following third chapter shifts the focus on problem analysis and dissects the core problems with real-time water testing. It enumerates the criteria which can be used to choose sensors, power management, environmental resilience, and the problems of data interpretation. The chapter contains an analysis of the components and tests the possibility of integration of a bottle-sized form. It also covers the technical and logistical choices made according to the issues identified during the first planning.

In the fourth chapter, the methodology and design of the system is given. It presents the entire block diagram of hardware architecture, then a flow diagram of the required steps execution to complete the system functions. This chapter will be split into two major sections of its design, hardware system design (microcontroller, sensors, display, battery, enclosure) and software system design (sensor data acquisition, AI model integration, user interface logic). It also describes the data gathering, training, testing and implementation of the AI component that would determine whether water is safe or unsafe with the input related to the sensor.

The fifth chapter dwells upon the development of tools and techniques. It describes the tools, libraries and platforms that will be used to program the hardware, to train machine learning models and to simulate. As an example, it can use Arduino IDE to develop firmware, TensorFlow Lite-based machine learning models can be run on an embedded system, and cloud platforms with optional web dashboards are used. It also addresses methodologies in testing water parameters, data logging as well as testing of sensor accuracy.

Lastly, a report that is a detailed conclusion is given in the sixth chapter. It goes back to the original goals, outlines the major project outputs, and mentions the performance of the system on what the intended goals were. The chapter also summarizes the restrictions which are present now, including calibration slippage, power execution, or sensor life, and provides an idea of future improvement. This can be better AI accuracy, waterproofing design, the support of additional water parameters, or similar connection with public health systems.

CHAPTER 2

LITERATURE REVIEW

Chapter 2: LITERATURE REVIEW

This chapter is an overview of available technologies, projects and available solutions regarding water quality monitoring. It contains two main parts: (1) Related Technologies, mentioning the enabling technologies supporting the creation of intelligent water quality checkers, and (2) Related Projects that review past work and commercial products in the area. The chapter ends with a view of shortcomings that have pervaded the current solutions and which this project tries to fill.

2.1 Related Technologies

In this section of the report, we will discuss the various technologies that are related to **HydroGuard**.

2.1.1 Multi-Parameter Water Quality Sensors

The current water quality testing relies on a variety of sensors that determine physical, chemical, biological parameters of water. The common environmental sensors include pH sensor, TDS (Total Dissolved Solids) sensor, turbidity sensor, temperature sensor, conductivity sensor, and the dissolved oxygen sensor.

Li et al. (2022) provides the information that these sensors may be integrated with microcontroller-based systems with their analog or digital data interfaces (I2C, UART) which makes it possible to obtain and process the data in real-time. These sensors must be precise and longstanding, which are critical in creating water safety assessments that people can rely on. Developments in MEMs technology now allow these to be small and cheap, they can thus be fitted into portable systems such as the proposed smart bottle [14], [23].

2.1.2 Artificial Intelligence in Water Analysis

Artificial intelligence is a game changer when it comes to reading sensor data. The currents of the raw water measure which can be quite inconsistent, and noisy rendering manual interpretation incomplete. Such researchers as Kumar et al. (2021) have suggested the application of machine learning (ML) models, including decision trees,

support vector machines (SVM), and artificial neural networks (ANN), to assort water quality on a scale of classes, including but not limited to safe or contaminated.

In addition, publicly available environmental databases or sets of labeled data by the government health agencies can be used as training ML models. The resulting trained model can be subsequently deployed to edge with TinyML frameworks such as TensorFlow Lite that enables inference on microcontrollers without cloud connection [13], [22].

2.1.3 Internet of Things (IoT) in Environmental Monitoring

IoT technologies have transformed the process of monitoring and sharing the environmental parameters. Commonly, in the smart water monitoring the sensors communicate with microcontrollers (e.g., ESP32 or Arduino) which relays information using Wi-Fi, Bluetooth, or LoRaWAN technologies to cloud-based services. Zhang et al. (2023) also states that IoT allows building distributed sensor networks that may send feedback to its users in real-time and officials.

The mobile apps and dashboards could be used privately to visualize the trend, memorize a warning, and keep the past data. In such a project, IoT integration produces such effects as the data concerning water quality could be visualized on a smartphone or the browser [19], [21].

2.1.4 Embedded Systems for Portable Testing

Small smart devices are at their core embodied in embedded systems. Microcontrollers like the ESP32 have in-built wireless modules, multiple analog inputs, and allow running of AI inference models. When combined with small displays (e.g. OLED or TFT) and battery packs they facilitate the creation of “field ready” tools that are self-powered.

Sensor compatibility, power efficiency and hardware flexibility of microcontrollers indirectly influence the structure of portable testing kits. The development is simplified with the use of open-source platforms such as the Arduino ecosystem and edge AI frameworks that allow local data processing and classification [6].

2.1.5 Cloud and Mobile Integration

Many contemporary environmental monitoring systems have integration or support apps to increase their availability and accessibility. The users can see real-time readings, trend graphs, see some historical changes and be alerted in case of unsafe readings. The two most popular frameworks to use cloud dashboards and IoT is Blynk, Thingspeak or Firebase. Custom web or mobile application interfaces are Flask and React Native.

In the given project, the connectivity feature will allow users to get AI-enhanced interpretations of water safety data remotely, which is in line with the idea of modern and comfortable user experience [2], [19].

2.2 Related Projects

2.2.1 IoT-Based Water Quality Monitoring System (2020)

Patel et al. presented a system that is based on the application of IoT to present a water monitoring system with the use of Arduino and a variety of sensors (pH, TDS, turbidity). Data was transmitted to ThingSpeak to monitor it in real time [15], [21]. Despite being applicable in fixed installations (such as water tanks), it did not have AI capabilities, portability, and the capability to view the display on device.

2.2.2 Smart Water Bottle for Athletes (2021)

In this student-led project at MIT, a smart bottle was involved, which monitors hydration through sensors of water consumption. Although a creative idea in fitness, it was not dealing with real contamination of water or even testing of quality. It demonstrates that electronic bottle electronics is possible and usable [26].

2.2.3 AI-Powered Water Safety Classifier (2022)

In Germany, researchers trained a neural network in water tests and installed it on a Raspberry Pi to label the water to be safe or not. The system was however bulky, needed

mains power and lacked any form factor that would allow it to be easily portable, or perform real-time tasks to a non-technical user [25].

2.2.4 Commercial Portable Test Kits (e.g., TestDrop Pro)

Various consumer products are marketed that make purported single-drop (usually through the electrical conductivity) tests of water quality. Nevertheless, they test mostly TDS, which cannot be considered a full-fledged level of water security. Few of them offer the AI interpretation and live feedback [17].

Table 1: Comparison of Related Works in Water Quality Monitoring

Project / Product	Sensors Used	Approx. Cost	Accuracy	Wireless Support	Portability
IoT-Based Water Quality Monitoring System (Patel et al., 2020)	pH, TDS, Turbidity	Low (~$50)	Medium (±5–10%)	Wi-Fi (ThingSpeak)	Low – Fixed installation
Smart Water Bottle for Athletes (MIT, 2021)	Flow / Hydration sensors	Medium (~$80)	High for hydration measurement	Bluetooth	High – Portable
AI-Powered Water Safety Classifier (Müller et al., 2022)	pH, TDS, Turbidity	High (~$150)	High (≥90%)	None	Low – Bulky, mains-powered
TestDrop Pro (Commercial)	Electrical Conductivity (EC)	Low (~$50)	Low – Only EC measurement	None	High – Handheld
Proposed System (Smart Water Quality Testing Bottle)	pH, TDS, Turbidity, Temp, Conductivity, DO	Low–Medium (~$80–100)	High (>90% with AI)	Wi-Fi / Bluetooth	High – Bottle-shaped, portable

2.3 Limitations/Problems in Literature Work

Though several research studies and commercial activities have been performed in water quality monitoring, upon critical analysis most of the currently existing system have left behind the complete range of the real-world applicability in terms of usability, intelligence as well as portability. Such constraints are as follows:

Narrow parameter coverage: most systems are based on one sensor (e.g. TDS or pH) and are insufficient to generate a full picture of water safety. This may result to erroneous findings and dangerous inferences.

Absence of AI Integration: Majority of the devices show raw convictions of the sensors, and they have no intelligent interpretation or safety classifications. This slows down using it by non-technical users and carries a low level of accuracy in making decisions.

Portability: Many of these academic prototypes are too big, power-consuming, or even dependent upon additional hardware to be practical everyday tools or in the field.

Weak Real-Time Feedback: The systems normally take cloud processing or manual analysis that increases response time and does not work under offline conditions.

Poor User Interface: In numerous designs, there are insufficient feedback (such as a display or a safety indicator), which makes them look garbled or not suitable to standard users.

Internet Dependency: The cloud reliant systems cannot be used in the rural areas or remote regions where testing of water is required the most.

Calibration and reliability concerns: Low-cost systems can be subject to sensor drift, and variation in calibration, but such issues have not been discussed in the literature.

Therefore, the proposed project consists of a multi-sensor bottle with compact size, AI functioning, offline operation, real-time safety classification, and the entire testing user-friendly experience in real life water testing.

CHAPTER 3

PROBLEM ANALYSIS

Chapter 3: PROBLEM ANALYSIS

3.1 Analysis of the Problem Specification

Clean and safe drinking water is one of the most debatable issues faced by the global society; water consumed by more than two billion people around the world does not meet the available safety measures estimated by the world health organization (WHO) [20]. Tainted water may contain toxins, germs and chemicals which cause severe health conditions like cholera, typhoid, diarrhea and other diseases transmitted through water. Although the technologies of water quality testing in laboratory conditions have been developed, there is still insufficient access to real-time, portable, and intelligent solutions to be used in daily or field testing. Testing in a laboratory setting may be accurate but it is among the most time-consuming processes, as it takes specialized equipment, personel, and time to perform, therefore, is not possible on a quick decision basis in a household, outdoor, or emergency setting.

There are incremental benefits in regard to accessibility on portable testing kits but they usually only consider one or two parameters e.g. pH or total dissolved solids (TDS) and provide raw results or crude color codes. These outputs need to be technically interpreted, and thus it is not to be used by a non-expert and more likely that they will be misinterpreted. Moreover, the majority of these devices do not make the intelligent data processing meaning that they do not aggregate various parameters into a single safety rating. Other limitations are the lack of real-time alerting, manual tracking of calibration, and in other situations lack of internet to process results which hinders their application at the local level in the countryside or offline areas.

The Smart Water Quality Testing Bottle is conceived as an intervention to address such gaps due to its combination of several calibrating sensors and an embedded AI-based decision-making system. The intended device utilizes essential parameters of pH, turbidity, TDS, conductivity, temperature, and also includes dissolved oxygen detection; heavy metals may be detected, as well. It will automatically categorize water in real-time based on an AI model trained on a variety of water quality data on board, so it works entirely offline. There will be an existing display that gives an immediate response, and wireless connection is optional whereby results can be logged and

analyzed using a mobile app. The housing will be small and moveable so that this device can be used in diverse settings, both home and military operations, without being restricted to this or that type of containers.

To realize such goals, the system will have to satisfy certain functional and non-functional requirements. Functionally, it should be able to do multi-parameter sensing, run AI algorithms to make immediate classification and show clear results on its user-friendly interface. Among the non-functional requirements, one may include portability, protection against the dust and splashes that the environment can be polluted by, long battery, and ability to work offline and in the online mode. Among the tools that will be used in the development include Arduino IDE and PlatformIO to program the firmware, TensorFlow Lite to deploy and use an AI model, and CAD software to design protective and ergonomic enclosure. Integration with cloud services such as firebase or Blynk can be used to manage logging of the data and creating alerts when connectivity to the internet is present. All these characteristics are meant to make the system technical, easy to use and fit to various situations that happen in the real world.

3.2 Required Components and Calibration

The Smart Water Quality Testing Bottle is also composed of both some software and hardware features which jointly come together so as to measure, process and present accurate results of the water quality.

Software wise, firmware development is possible through the Arduino IDE and PlatformIO allowing a multi-sensor and multi-input programming platform on the ESP32 microcontroller. This artificial intelligence model is developed in Python with TensorFlow/Keras and converted in to TensorFlow Lite for Microcontrollers to make the model run effectively on the ESP32 with minimal memory requirements. Optional connections are provided by means of using either Firebase or Blynk services, connected logs, notifications, and historical monitoring can be reified by use of an internet connection. With Figma, user interface designs of the integrated OLED display and the mobile application are designed in a manner that information is well presented and friendly.

On the software front, the system core is an ESP32-WROOM microcontroller, which is a dual-core, and 240 MHz is capable of supporting integrated Wi-Fi (802.11 b/g/n) and Bluetooth 4.2 [5]. The module was selected because it supports multiple sensors and supports processing AI inference and wireless connectivity, all without having to add additional modules. The sensing array contains:

- pH Sensor (SEN0161-V2) A sensor that measures acidity/alkalinity of the material provided based on a range of 0-14 pH and accuracy of measuring pH is +- 0.1 pH [3].
- Turbidity Sensor (SEN0189) -Measures clarity of water between 0 and 1000 NTU and has ± 5 accuracy [4].
- Total Dissolved Solids (TDS) Sensor - Detects dissolved solids with 0-1000 ppm patterns and an accuracy of +/-10 per cent.
- Electrical Conductivity Sensor (DFR0300) Ionic content 0 to 20 mS/cm with an accuracy of ±2%.
- Temperature Sensor (DS18B20) 12-bit; temperature range of between -55 to 125 C and an accuracy of +/? 0.5 degrees C.
- Dissolved Oxygen (DO) Sensor- Monitors the amount of oxygen present with a range between 0-20 mg/L and a net accuracy of 0.2 mg/L [14].
- Optional Heavy Metal Sensor Electrochemical sensor to detect metal, lead or copper.

The sensors connect to ESP32 in a combination of analog connections (pH, turbidity, TDS, conductivity, DO, heavy metals), I2C to the OLED display, and 1-Wire to the temperature sensor. It has a 0.96″ OLED screen (128×64 resolution) to display both the numerical values and AI safety classifications ahead of time on the device. The power is provided with 2000 3000 mAh rechargeable Lithium-ion battery with a charge management module to ensure safety.

Calibration will make the device produce correct and identical measurements. Standard buffer solutions of pH 10.0, 7.0, and 4 and used in calibrating the pH sensor. Turbidity sensor is compared with Formazin reference standards at various NTU values. TDS and

conductivity signals are calibrated by means of KCl of known concentration. The process of calibration of the DO sensor involves the use of water that has oxygen in 100% and sodium sulfite in 0. The temperature sensor is calibrated before the thermometers found in the laboratory. The calibration coefficients are saved in a non-volatile memory in order to maintain the device sensitivity to the passage of time after which measures are given to recalibrate in case drift is observed [18], [23].

3.3 Decision Making based on Problem Analysis

The ergonomics of the Smart Water Quality Testing Bottle was driven by the requirements to eliminating the particular drawbacks of the other technologies on the water testing market as well as maintaining the system small, efficient, and simple to carry. The choice of Espressif ESP32-WROOM microcontroller was a tactical decision since it has the computing power to process data across multiple sensors at once, has wireless connectivity available to use to implement optional remote monitoring capability, and can run an on-device embedded AI model locally without having to resort to cloud processing. This would guarantee the ability of the device to perform well both on connected and offline operations, which is effective when the setting may be in a remote locality, disaster response areas, or even outdoor trips that can not have internet around them.

The selection of a multi-sensor method as opposed to using a single parameter of measurement was important to increase the accuracy and reliability of the findings. Single parameters like pH, turbidity, TDS, conductivity, temperature, dissolved oxygen, and heavy metals will give only partial answers concerning the quality of water but when they are analyzed together in the process of AI analysis, results are always more reliable. Such an integrated solution removes the issue of false confidence which may arise when single or few water quality indicators are monitored.

Embedded AI/ML model supports the possibility to process the readings of all sensors in real-time and classify water as fit or unfit according to the WHO criteria. This removes the reliance upon the user to make his/her own interpretation of raw measurements, thus removing the likelihood of error, particularly amongst non expert operators. The choice of offline AI operation instead of cloud computing processing of

information makes the system operable in any location without concern of data privacy, and optional online capabilities including data logging and analysis of trends over time is possible when connected.

Mechanically, the enclosure should be made small, enclosed, portable, and enable steady sensor installation to get a measured reading. It is not bound to a specific container shape since design can be adapted easily to variations in deployment scenarios and the design does not lock a project to a particular design by forcing the project to adopt the specific structure. Wiring and locations of sensors are carefully considered internally in order to avoid electrical interferences and cross-contamination of measurements. A lithium-ion battery with ability to be recharged was selected because it gives longer periods of operation and use of low-power modes of firmware consumption when the device is idle will assist in saving energy.

The risk analysis was used to determine any possible vulnerability. Time drift in sensors is compensated by the periodicity of calibration prompts and the calibration data is saved in non-volatile memory to allow the effectiveness of accuracy over the period of time between uses. The risk of the battery becoming empty is decreased with power saving modes and a battery icon level in the display. The risk of AI misclassification is mitigated through a training set with a diverse large dataset and having a confidence score in the results that will trigger a retest should the confidence be on the lower end. The environmental protection measures against destruction can be computerized through splash resistance sealing, protective gasket, and conformality coating electronic appliances.

Table 2: Comparison between Traditional and Modern Monitoring

Feature	Traditional Portable Kits	Smart Water Quality Testing Bottle
Parameters Measured	Typically 1–2 (e.g., pH, TDS)	6–7 including pH, turbidity, TDS, conductivity, temperature, DO, heavy metals
Result Interpretation	Manual, requires expertise	Automatic AI-based classification
Operation Mode	Fully offline	Offline with optional online data logging
Real-Time Alerts	Not available	Yes, on-device safe/unsafe indication
Calibration Tracking	Manual, user-dependent	Automatic reminders and stored calibration data
Data Logging	Rarely available	On-device storage and optional cloud sync
Usability	Risk of error for non-experts	Designed for both experts and general users

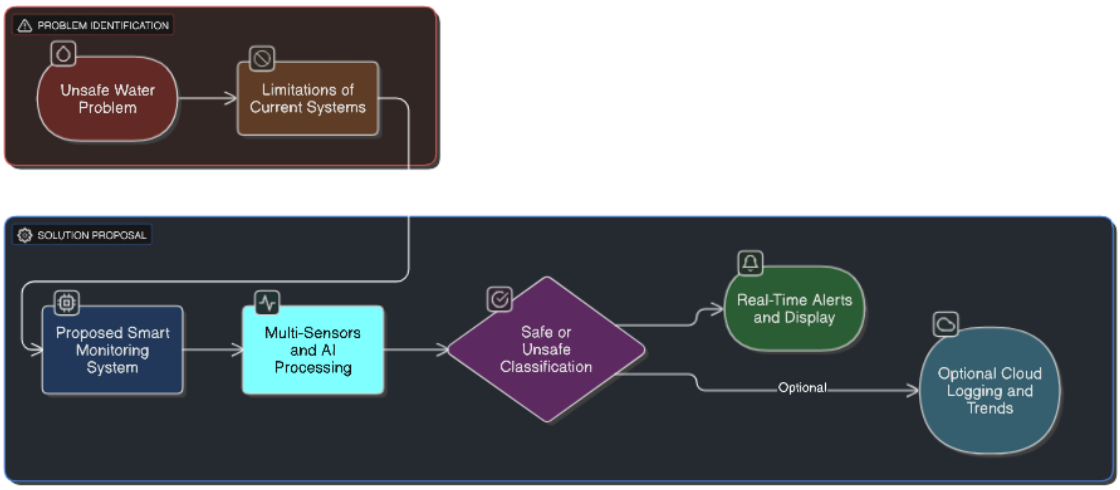


Figure 1 - Problem identification and proposed solution workflow for the Smart Water Quality Testing Bottle.

CHAPTER 4
PROJECT DESIGN AND METHODOLOGY

Chapter 4: DESIGN AND METHODOLOGY

4.1 Overall Project Design

The Smart Water Quality Testing Bottle is developed as a built-in system to consist of several calibrated sensors, an ESP32 microcontroller, and embedded AI process to enable a real-time evaluation of the drinking water safety. The design is portable, offline capable, and user-friendly, hence can be used in different contexts including monitoring at home, undertaking activities outdoors, and making field checks in remote locations.

The centering point of the system is the ESP32-WROOM microcontroller to receive six to seven sensors reading pH, turbidity, total dissolved solids (TDS), electrical conductivity, temperature, dissolved oxygen (DO) and, if necessary, heavy metals. Their readings are processed before they are read to eliminate any temperature effect, eliminate any noise and normalize the data. The ESP32 is directly running an AI/ML model-trained on a dataset of water samples, with TensorFlow Lite for Microcontrollers providing real-time water classification locally as being either Safe or Unsafe by WHO standards [20].

The outcomes appear in real time on an OLED 0.96-inch screen and the results will feature a visual or audible alert, should it prove unsafe. Non-obligatory Bluetooth or Wi-Fi can be designed to access the device to sync the results with a mobile application in order to track history, provide graphical analysis, and reminders that the calibration is due. A rechargeable lithium-ion battery powers it, and an onboard charging module controls it to be used in the field longer.

4.1.1 Block Diagram

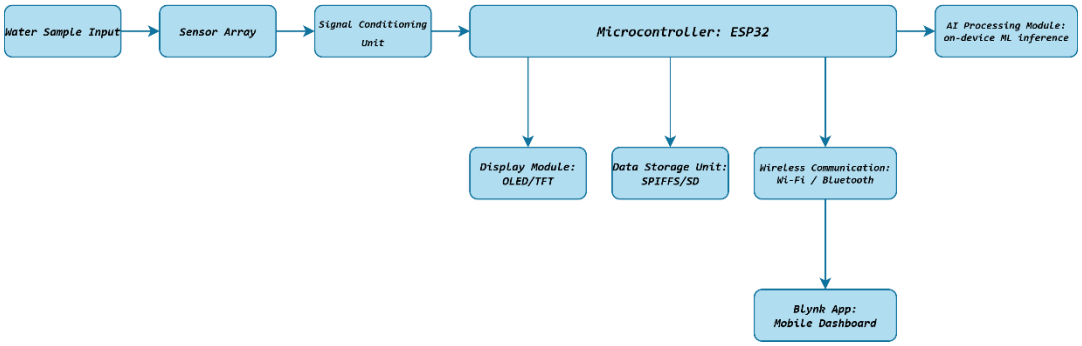


Figure 2 - Block diagram describing the functionality of the system.

Figure 4.1.1 is the block diagram of the overall functionality of the Smart Water Quality Bottle. The core part of the system is the ESP32 microcontroller which organizes all the actions. It starts by the user gathering a sample of water in the supplied inbuilt container. When the sample has been identified, the ESP32 turns on the sensor array that is comprised of pH, turbidity, TDS, conductivity, temperature, dissolved oxygen, and heavy metals sensors. Raw sensor data is recorded, sent to the preprocessing module, and here noise is filtered, values are normalized and temperature compensation performed. The smoother data is then inputted to the AI-based classification feature that decides on whether the sample is in the set safety walls or not. The outcomes of the classification will be illustrated on the OLED screen and sent through Bluetooth or Wi-Fi to the corresponding cell phone application of the user. Meanwhile, readings are kept locally and can be uploaded (optionally) to a cloud database where they are tracked on an ongoing basis. Peripheral modules, including charging, power management circuits and optional calibration guidance interface are also depicted in the block diagram to provide reliable, energy efficient operation.

4.1.2 Flow Diagram

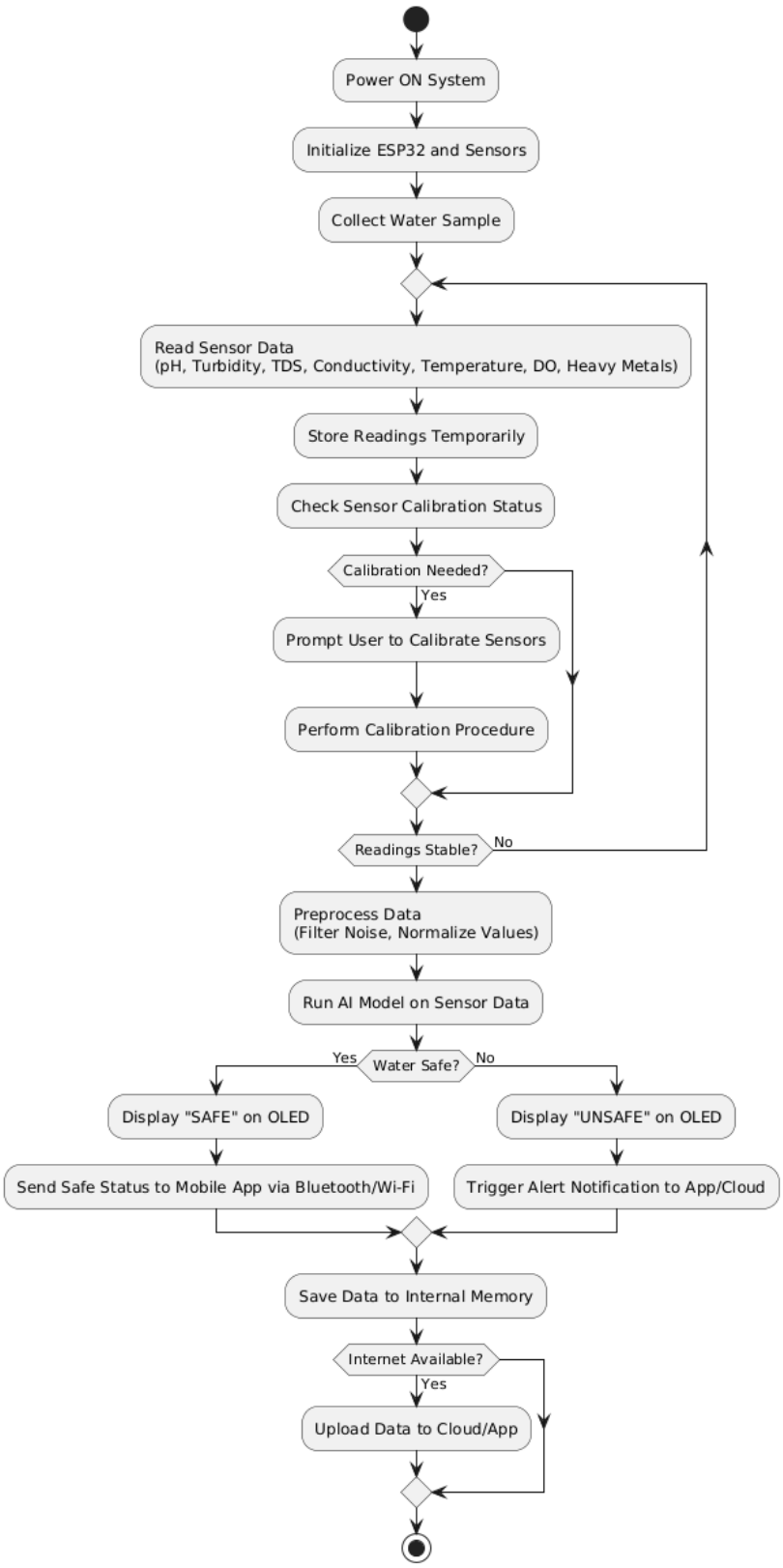


Figure 3 - Flow Diagram

As the flowchart drawn in Figure 4.1.2 shows, the process cycle of the Smart Water Quality Bottle represents the chronological order of its work, starting with initialization and ending with the delivery of final results. When powered up, all hardware on the ESP32 is initialized, this includes the sensor array, display module and wireless communication interface. The user is then asked to sample and store a water sample by the system. The sensor readings are taken serially with a stability check done to maintain consistency of the measurements. In case of calibration, the user is taken through the procedure until he reaches stable readings. The preprocessing is done to eliminate noise, exclude outliers, normalize data and correct the readings to the impacts of the temperature. The cleaned data is subjected to analysis, being fed to the onboard AI model, which decides which category of water to use: Safe or Unsafe. The outcome is displayed on the OLED screen, and unsafe findings cause instant warnings to the related mobile application. The readings and the classification of all readings are stored on non volatile memory then sign in to the cloud when and where there is internet and monitoring done remotely. The device functions and operates correctly, efficiently, and in real-time because of this structured flow; thus investors get results of credible water safety.

4.2 Methodology

This section describes the complete implementation process of the Smart Water Quality Bottle, from sample collection to budget planning. The methodology ensures accurate water quality analysis, user-friendly operation, and cost-effective deployment.

4.2.1 Sample Collection

It starts with the obtaining of a water sample in the built-in container. The design of bottle removes the chances of contamination during sampling as the bottle is designed with a sealed cap and pre-washed inside. The sampling can either be triggered manually through a push button or pre set intervals automatically as dictated by the mode of operation.

4.2.2 Sensor Data Acquisition

When the samples are detected, ESP32 microcontroller turns on the built-in sensor array that contains:

- PH Sensor- acidity/alkalinity [3].
- Turbidity Sensor – detection of water clarity [4].
- TDS Sensor - detection of total dissolved solids [14].
- Conductivity Sensor-sensor to with ionic concentration [14].
- Temperature Sensor - to read and to compensate on temperature [14].
- Dissolved Oxygen Sensor: – saturation of the oxygen [14].
- Heavy Metal Sensor-detection of lead and mercury and other contaminants [14].

Resulting values of the sensors are read serially by ESP32 [5] and stored in volatile memory where they undergo preprocessing.

4.2.3 Calibration Process

Calibration of the sensors is necessary within an interval. The system checks stored calibration data at the startup. In case the calibration is necessary, the OLED display and the mobile app take the customer through regular procedures with the help of known solutions of calibration [18]. They also include the calibration parameters which are saved in EEPROM to survive reboots.

4.2.4 Data Preprocessing

Data are collected and then passed through some steps of data processing prior to analysis:

- Noise Filtering: Stabilising readings through moving averages.
- Outlier Removal:Disregarding outlying values outside acceptable limits.
- Normalization: reducing the values to the expected range of the AI model.
- Temperature Compensation: The pH and conductivity are adjusted according to measured temperature [13], [22].

4.2.5 AI-Based Classification

The AI model, which is embedded on the ESP32 or in the cloud, conducts an evaluation of the pre-processed data [7]. The sample has been modeled as follows:

- Safe- any reading below the acceptable levels.
- Unsafe -There are one or more parameters that are not within safety limits.

The thresholds are in line with the international standards (e.g., WHO, EPA) [18] and able to be revised in the mobile application.

4.2.6 Display and Notification

The OLED screen shows results:

- SAFE in green for acceptable water quality.
- UNSAFE in red for contaminated water.

Simultaneously, the results are posted using Bluetooth or Wi-Fi to the mobile application [2]. Unsafe values make the immediate display of detailed readings and timestamps.

4.2.7 Data Logging and Cloud Upload

Internal non-volatile memory stores all the results. Data with internet access is uploaded to Firebase where it can be stored long-term and analysed historically and accessed remotely [7].

4.2.8 Power Management

Low power modes are utilised by the system to increase battery life. The ESP32 will go to deep sleep between measurements, and when idle any non-essential peripherals are powered down [5]. There is charging that is controlled by a TP4056 charging module through USB.

4.2.9 Testing and Validation

Review of validation encompasses lab testing under strict conditions as well as field promotion under the condition of tap, bottled and natural water. Important indicators are:

- Accuracy -accuracy against laboratory test results.
- Precision & Recall – accuracy of unsafe finding.
- Response Time: time between sample input and result displa

4.2.10 Budget Analysis

4.2.10.1 Hardware Components Cost

Table 3: Hardware Components Cost

Component	Qty	Unit Price (AED)	Total (AED)
ESP32 Microcontroller	1	~35	~35
pH Sensor Module	1	~55	~55
Turbidity Sensor	1	~35	~35
TDS Sensor	1	~40	~40
Conductivity Sensor	1	~75	~75
Dissolved Oxygen Sensor	1	~120	~120
Heavy Metal Sensor	1	~150	~150
OLED Display (0.96")	1	~25	~25
Lithium-Ion Battery	1	~45	~45
Charging Module (TP4056)	1	~10	~10
Bottle Housing + Waterproofing	1	~50	~50
Wiring & Connectors	-	~20	~20
Total Hardware Cost			~660 AED

4.2.10.2 Software & Cloud Services Cost

Table 4: Software & Cloud Services Cost

Item	Price (AED)
Blynk Cloud (Free Tier)	**0**
AI Model Training (Google Colab Pro)	**45**
Firebase Cloud Storage (Free Tier)	**0**
Total Software Cost	**45 AED**

4.2.10.3 Overall Project Cost

- Total Estimated Cost: 705 AED
- Contingency (10%): ~70 AED
- Final Budget: ~775 AED

4.2.10.4 Feasibility

Free-tier cloud services and the use of open-source software make the project cost-efficient [1], [2]. Its modular construction enables it to upgrade only sensors and not the whole system thus lowering the operation costs long-term.

4.3 Implementation Timeline

The project envisioned in a time span of 13 weeks, wherein there will be phases that will be dedicated to design, development, testing and deployment processes. The breakdown is systematic and gives enough room to repeat on the basis of the test results.

Table 5: Implementation timeline table

Duration	Task	Timeline
1 week	Requirements Analysis	Week 1
1 weeks	System Design	Week 2
1 weeks	Procurement	Week 3
2 weeks	Hardware Assembly	Week 4-5
2 weeks	Firmware Development	Week 6-7
2 weeks	AI Model Development	Week 8-9
1 week	Mobile App Integration	Week 10
1 week	Testing & Calibration	Week 11
1 week	Field Testing	Week 12
1 week	Documentation & Final Review	Week 13

Ongoing Maintenance:

The Smart Water Quality Bottle will need to be maintained regularly after being deployed, in order to maintain accuracy and reliability in its results:

- Sensor Calibration: Standard solutions every 24 weeks according to the frequency of utilization.
- Battery Checks: Check battery and charge once a month; every 12-18 months replace with new battery.
- Firmware Updates: Push via USB or OTA (Over-the-Air) improvements to the AI accuracy and features.
- Data Backup: Back up data with the cloud constantly to stop any losses.
- Physical Inspection: After long service, this can be checked by looking at cracks that might be presented in the houses, seal check and water proofing.

4.4 Summary

In this chapter, the design, methodology, and implementation plan of Smart Water Quality Bottle were described. Section 4.1 has introduced the architecture of such system with the use of block diagrams and flowcharts to show the interactions between hardware and software component to provide real-time water safety analysis. In section 4.2, the entire methodology of water sampling, data capture of sensors, AI-assisted water quality classification, notification of the user and budgets was explained in detail. Section 4.3 contains a 13 week timeline of the implementation, a Gantt chart as well as the maintenance strategies to be introduced to ensure that accuracy and usability of the system are maintained. The intended structure guarantees a harmony between technological proficiency, economic viability, and sustainability further producing the possibility of successful implementation under a broad range of real life environments.

CHAPTER 5

TOOLS AND TECHNIQUES

Chapter 5: TOOLS AND TECHNIQUES

5.1 Hardware & Technical Specifications of Components

The hardware components used in this project are described below, along with their technical specifications:

- **ESP32-WROOM-32**



Figure 4 - ESP32-WROOM-32

It is Wi-Fi/Bluetooth dual-core microcontroller that acts as a central processing unit of the system and is responsible of multi-sensor data acquisition, inference based on AI, and wireless transfer of data to the mobile application.

Specifications:

- Tensilica Xtensa LX6 Dual-Core 32-bit, maximum of 240 MHz
- Operating Voltage: 3.0V- 3.6V (recommended 3.3V)
- Wi-Fi Standards: IEEE 802 Eleven.b/G/N (2.4 GHz)
- Bluetooth: V4.2 BR/EDR and BLE
- Flash Memory: 4 MB(external)
- GPIO: 34 general purpose pins (ADC, DAC, UART, SPI, I2C, PWM)

- **pH Sensor (SEN0161-V2)**



Figure 5 - pH Sensor (SEN0161-V2)

Used to measure the acidity or alkalinity of the water sample in real-time for detecting harmful imbalances.

Specifications:

- Measuring Range: 0 – 14 pH
- Accuracy: ±0.1 pH
- Operating Voltage: 3.3V – 5V
- IV. Output: Analog signal

- **Turbidity Sensor (SEN0189)**

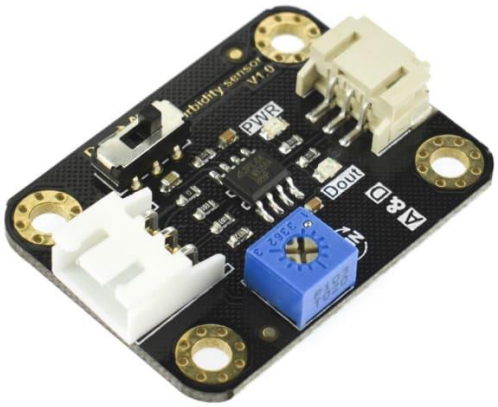


Figure 6 - Turbidity Sensor (SEN0189)

Detects water clarity by measuring the presence of suspended particles or sediments.

Specifications:

- Measuring Range: 0 – 1000 NTU
- Accuracy: ±5% F.S.
- Operating Voltage: 5V
- Output: Analog voltage proportional to turbidity

- **OLED Display (0.96", 128×64 pixels)**

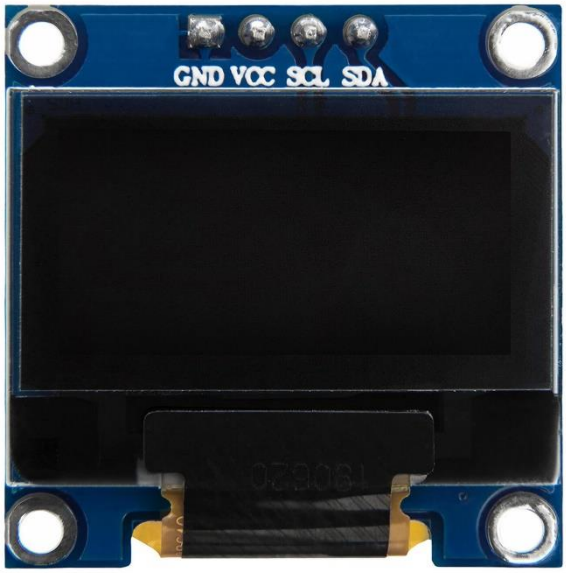


Figure 7 - OLED Display (0.96", 128×64 pixels)

Displays water parameter readings, AI-based classification results, and system status.

Specifications:

- Display Type: Monochrome OLED
- Resolution: 128 × 64 pixels
- Operating Voltage: 3.3V – 5V
- Interface: I²C

- **Battery**



Figure 8 - Battery

Provides portable power to the entire system.

Specifications:

- Type: Lithium-ion
- Voltage: 5V

- **Jumper Wires**

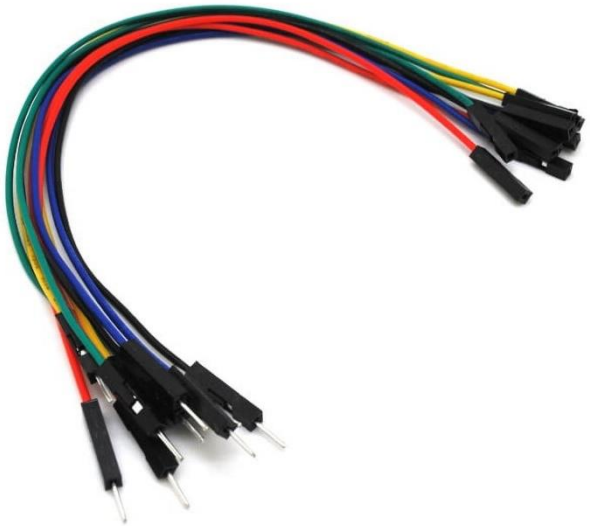


Figure 9 - Jumper wires

Connects remote electric circuits used for printed circuit boards

Specifications:

- Types: Male-to-male, female-to-female, male-to-female
- Length: 10cm–30cm

5.2 Software and Simulations Tools

The systems under development and simulation are accompanied by the following software tools:

Arduino IDE [1]

Programming of ESP32 microcontroller is based mostly on the Arduino Integrated Development Environment (IDE). It offers direct accessibility to write, compile, and upload firmware in charge of gathering information sensed, communicating with a wireless connection, and administering the qualification of water classes with the help of AI. Third party libraries can easily be integrated into the environment as well (sensor drivers, communication protocols, etc.) and TensorFlow Lite [7], which allows seamless development and deployment.

Blynk App [2]

The system has a mobile interface provided by the Blynk IoT platform that makes possible tracking of the water quality parameters in real-time and managing them remotely. The smartphone-based app allows users to monitor real-time values of the sensors on the dashboard (which can be customized), monitor historical trends, and also sends push notifications automatically when a parameter exceeds a threshold. This makes the user aware instantly and can interact with the system located remotely without involving immediate contact with the hardware.

Proteus Simulation Software [9]

Proteus is more like a tool to simulate the electronic circuit design before enacting it physically. It facilitates emulation of ESP32 and related connected sensors as well as display modules, proper pin assignments, power compatibility and communication protocols are all performed in this simulation. With the test of the design in a hypothetical environment, any challenges of hardware integration in the making are produced and solved before the actual development is successful.

Google Colab [7]

Machine learning model development and training is done using Google Colab, which utilizes its cloud-based GPU acceleration. It is designed to pre-process any water quality dataset, to train and optimize classification models and deploy them to resource-limited devices. These last models are exported to TensorFlow Lite format to execute fast on the ESP32 without an internet connection thereby performing on-device AI inference.

5.3 Mobile Application Features

The Smart Water Quality Testing Bottle mobile application is designed on the Blynk IoT platform [2], which will offer full interaction with the interface when monitoring and controlling the system. A personalized dashboard in the app provides real time values of water quality parameters such as pH, turbidity, total dissolved solids (TDS), conductivity, dissolved oxygen and temperature. Historical data logging allows users to capture the relationship on long-term water quality changes in graphical form, whereas the results of classification based on AI calculations display the condition of the water as safe or unsafe in an agreed upon easy-to-understand form. Threshold-based push notifications are automatically issued when sensor data happens to surpass pre-configured limits, as it guarantees the prompt indication of dangerous situations. Remote calibration commands are also applicable in the application where it is possible to start the process of sensor calibration but without physically touching the device. Data caching will allow offline reading so that any reading taken when there is a communication outage is automatically updated when the connection is restored and continued monitoring is consistent and reliable.

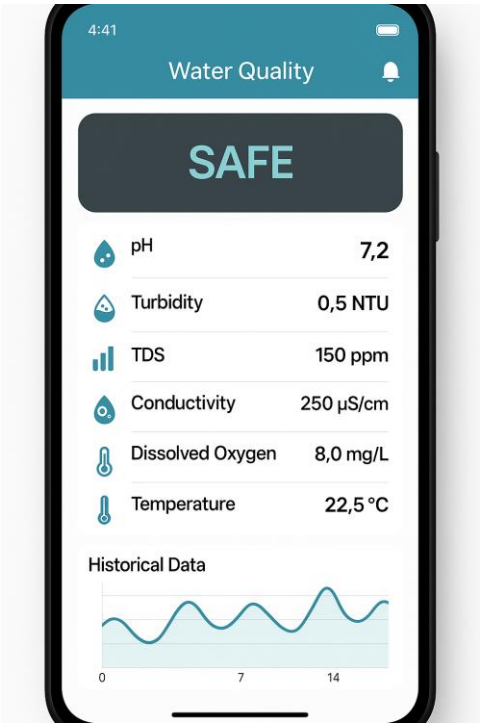


Figure 10 - App Mockup

CHAPTER 6
CONCLUSION

Chapter 6: CONCLUSION

The Smart Water Quality Testing Bottle effectively proves the possibility of combining a wide array of sensors to measure water quality, inert embedded artificial intelligence to label their output, and cellular, low-power, and low bandwidth connect to the mobile IoT in one portable device. The novel project resolved major limitations all through the design implementation process, which included sensor calibration, power management, environmental protection and real time data interpretation. Thanks to the AI inference capabilities of the ESP32 microcontroller, the system can work adqtably both online and offline, increasing the scope of its application to remote and low-connectivity regions.

The mobile application is also present, which increases the ease of access as users can get alerts in real-time, access previous data dynamics, and remotely control calibration. The prototype achieves its most important goals of offering precise, fast, and user-friendly water safety tools of measurement and is also in line with the United Nations Sustainable Development Goals (SDG 3, SDG 6, and SDG 9).

Even though the prototype fulfills the expectation of the design, some developments can be improved. Future studies could include training the AI models on larger, more comprehensive datasets, addition of the detection of heavy metals, smoothing out the waterproofing to work with harsher conditions, and the option of charging using solar energy to be more sustainable. The expanded utilization of this technology could also support dramatic increases in the health outcomes of the general population, especially in underserved populations and cases of emergencies response.

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